Classical Mechanics 1: Quiz 1

1. **[Taylor 1.11 ⋆]** The position of a moving particle is given as a function of time t to be

$$\mathbf{r}(t) = \hat{\mathbf{x}}\, b\cos(\omega t) + \hat{\mathbf{y}}\, c\sin(\omega t),$$

where b, c and ω are constants. Describe the particle's orbit.

2. **[Taylor 1.39 ⋆⋆]** A ball is thrown with initial speed v_0 up an inclined plane. The plane is inclined at an angle ϕ above the horizontal, and the ball's initial velocity is at an angle θ above the plane. Choose axes with x measured up the slope, y normal to the slope, and z across it. Write down Newton's second law using these axes and find the ball's position as a function of time. Show that the ball lands a distance

$$R = \frac{2v_0^2 \sin\theta\cos(\theta+\phi)}{g\cos^2\phi}$$

from its launch point. Show that for given v_0 and ϕ, the maximum possible range up the inclined plane is

$$R_{\max} = \frac{v_0^2}{g(1+\sin\phi)}.$$